Practice workbook · Connected rates: linking two changing quantities

Using a known rate of one quantity (for example volume) to find the rate of another (for example surface area) through a shared variable.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-15

1. Question 9 · part whole · 7 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part whole only.

9 A cube has edges of length x cm.

The total surface area, A cm^2, of the cube is increasing at a constant rate of 0.45 cm^2/s

Find the rate of increase, in cm^3/s, of the volume of the cube at the instant when the total surface area of the cube is 384 cm^2

(7)

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1. Question 9 · part whole · 7 marks

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Worked-example retry · Attempt part whole only.

Question 9 continued

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(Total for Question 9 is 7 marks)

1. Question 9 · part whole · 7 marks · Working and answer

Working and answer

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2. Question 10 · part c · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

10

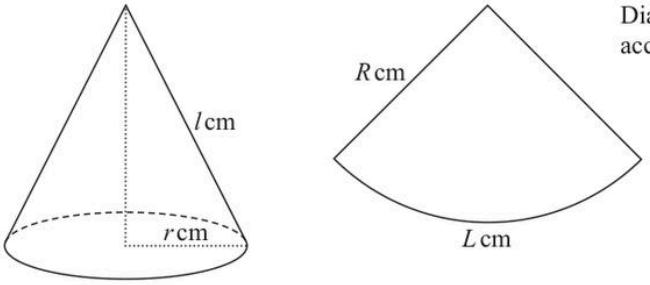


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right circular cone with base radius r cm and slant height l cm.
Figure 4 also shows a sector of a circle with radius R cm and arc length L cm.

The area of the curved surface of the cone is A cm^2

By considering how the sector of the circle can be folded to exactly form the curved surface of the cone with R and L suitably chosen,

(a) prove that $A = \pi rl$

(4)

Sand is poured onto a horizontal surface at a constant rate of 1.5 cm^3/s.
The sand forms a pile in the shape of a right circular cone with its base on the surface.
The curved surface area of the cone, A cm^2, increases in such a way that the height of the cone is always three times the radius of the base of the cone.

Given that $\frac{dA}{dr} = k\pi r$, where k is a constant,

(b) find the exact value of k.

(3)

(c) Calculate the rate, in cm^2/s, to 3 significant figures, at which the curved surface area of the pile is increasing when the height of the pile is 24 cm.

(5)

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P 6 2 2 8 0 A 0 2 6 3 2

2. Question 10 · part c · 5 marks

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Further practice · Attempt part c only.

Question 10 continued

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2. Question 10 · part c · 5 marks

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Further practice · Attempt part c only.

Question 10 continued

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2. Question 10 · part c · 5 marks

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Further practice · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 12 marks)

2. Question 10 · part c · 5 marks · Working and answer

Working and answer

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3. Question 6 · part b · 8 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

6

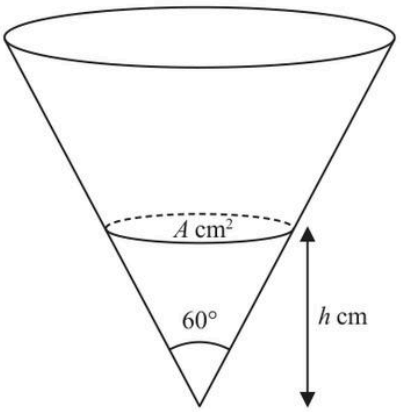


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a container in the shape of a right circular cone.
The container is fixed with its axis of symmetry vertical.
The vertical angle of the container is 60° as shown in the diagram.

At time t seconds, $t > 0$, the height of oil in the container is h cm and the volume of oil in the container is V cm^3

(a) Show that $V = \frac{1}{9}\pi h^3$

(3)

At time t seconds the surface area of oil in the container is A cm^2, as shown in Figure 4

Oil is dripping out of the bottom of the container at a constant rate of 4 cm^3/s.

(b) Find the exact rate of change, in cm^2/s, of the surface area of oil in the container when $h = 24$

(8)

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P 7 4 0 9 8 A 0 1 6 3 2

3. Question 6 · part b · 8 marks

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Further practice · Attempt part b only.

Question 6 continued

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3. Question 6 · part b · 8 marks

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Further practice · Attempt part b only.

Question 6 continued

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3. Question 6 · part b · 8 marks

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Further practice · Attempt part b only.

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Question 6 continued

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(Total for Question 6 is 11 marks)

3. Question 6 · part b · 8 marks · Working and answer

Working and answer

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4. Question 9 · part b · 8 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

9

Diagram **NOT** accurately drawn

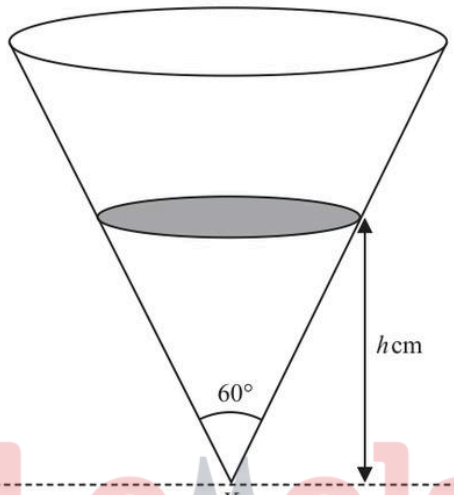


Figure 2

Figure 2 shows a hollow right circular cone fixed with its axis of symmetry vertical.

The vertical angle of the cone is 60°
Initially the cone is empty.

At time $t = 0$ liquid starts to fill the cone at a constant rate of $0.2\,\text{cm}^3/\text{s}$
At time t seconds after the liquid starts to fill the cone, the height of the liquid is h cm above X

(a) Show that $h = \sqrt[3]{\frac{9t}{5\pi}}$

(5)

The surface area of the liquid, shown shaded in Figure 2, is increasing at a constant rate of $p\,\text{cm}^2/\text{s}$ when $t = 6$

(b) Find, to 3 significant figures, the value of p

(8)

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4. Question 9 · part b · 8 marks

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Further practice · Attempt part b only.

Question 9 continued

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4. Question 9 · part b · 8 marks

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Further practice · Attempt part b only.

Question 9 continued

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4. Question 9 · part b · 8 marks

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Further practice · Attempt part b only.

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Question 9 continued

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(Total for Question 9 is 13 marks)

4. Question 9 · part b · 8 marks · Working and answer

Working and answer

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5. Question 9 · part b · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

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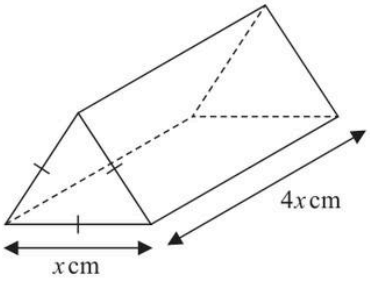


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a metal solid S.

The solid is a right triangular prism.

The cross section of S is an equilateral triangle with sides of length x cm.
The length of S is $4x$ cm.

The prism is being heated so that the cross sectional area is increasing at a constant rate of $0.03\,\text{cm}^2/\text{s}$.

(a) Find, giving your answer to 3 significant figures, $\frac{dx}{dt}$ when $x = 2$ **(5)**

(b) Find the rate of increase, in cm^3/s, of the volume of S when $x = 2$ **(3)**

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5. Question 9 · part b · 3 marks

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Further practice · Attempt part b only.

Question 9 continued

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5. Question 9 · part b · 3 marks

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Further practice · Attempt part b only.

Question 9 continued

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5. Question 9 · part b · 3 marks

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Further practice · Attempt part b only.



Question 9 continued

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(Total for Question 9 is 8 marks)

5. Question 9 · part b · 3 marks · Working and answer

Working and answer

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6. Question 10 · part whole · 9 marks

2020 JAN · Paper 02

Further practice · Attempt part whole only.

10 The volume of a sphere is increasing at a constant rate of $40\,cm^3/s$.

Find the rate of increase, in cm^2/s, of the surface area of the sphere at the instant when the radius is 4 cm.

(9)

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6. Question 10 · part whole · 9 marks

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Further practice · Attempt part whole only.

Question 10 continued

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(Total for Question 10 is 9 marks)

6. Question 10 · part whole · 9 marks · Working and answer

Working and answer

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7. Question 7 · part b · 5 marks

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Further practice · Attempt part b only.

7 The length of each side of a cube S_1 is increasing at a constant rate of 0.1 m/s.

(a) Find, in m^3/s, the rate of increase of the volume of the cube S_1 when the length of each side of the cube is 2 m.

(4)

The total surface area of a different cube S_2 is increasing at a constant rate of $0.05\,m^2/s$.

(b) Find in m^3/s, the rate of increase of the volume of the cube S_2 when the length of each side of the cube is 6 m.

(5)

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7. Question 7 · part b · 5 marks

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Further practice · Attempt part b only.

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Question 7 continued

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7. Question 7 · part b · 5 marks

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Further practice · Attempt part b only.

Question 7 continued

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7. Question 7 · part b · 5 marks

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Further practice · Attempt part b only.

Question 7 continued

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(Total for Question 7 is 9 marks)

7. Question 7 · part b · 5 marks · Working and answer

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